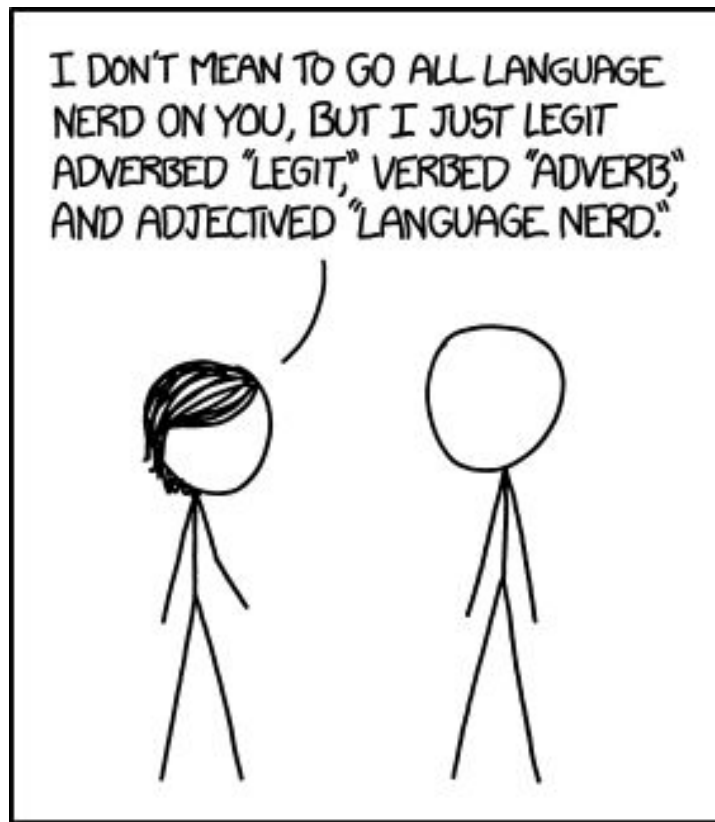


CS 159 Week 5:

Part-of-speech tagging



(<https://xkcd.com/1443>)

Two weeks ago:

A **language model** scores the probability of a given word sequence:

$$P(w_1, w_2, w_3, \dots, w_k)$$

The simplest language model is the **bag of words model**, which assumes words are completely independent of each other:

$$P(w_1, w_2, w_3, \dots, w_k) \approx P(w_1)P(w_2)P(w_3)\dots P(w_k)$$

Each $P(w)$ can be estimated as the **frequency** of word type w

Two weeks ago:

For a slightly more advanced language model, we can replace the independence assumption with the **Markov assumption**.

The Markov assumption states that the likelihood of a word depends only on a fixed **context window** of previous words.

For example, a **bigram** language model assumes that the likelihood of a word depends only on the previous word:

$$P(w_1, w_2, w_3, \dots, w_k) \approx P(w_1)P(w_2|w_1)P(w_3|w_2)\dots P(w_k|w_{k-1})$$

Visualizing a Markov language model

“Colorless green ideas sleep furiously.”



Visualizing a Markov language model

“Colorless green ideas sleep furiously.”



Colorless

green

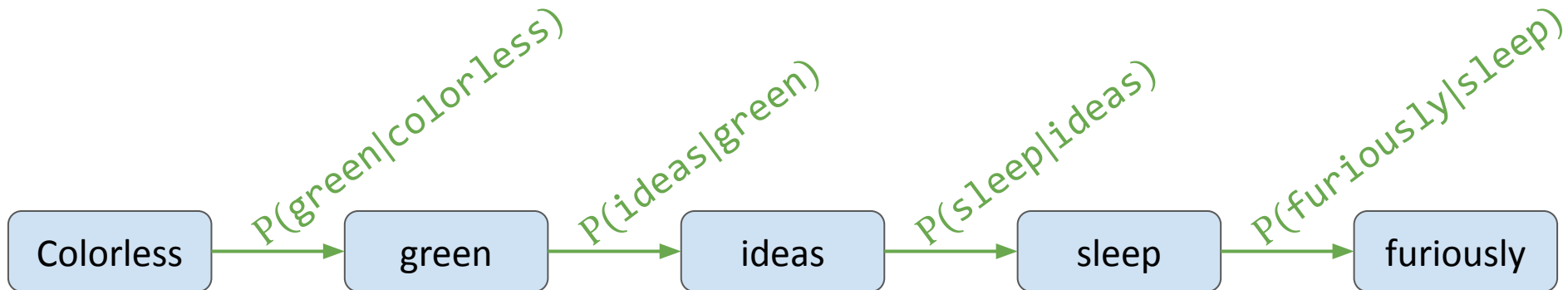
ideas

sleep

furiously

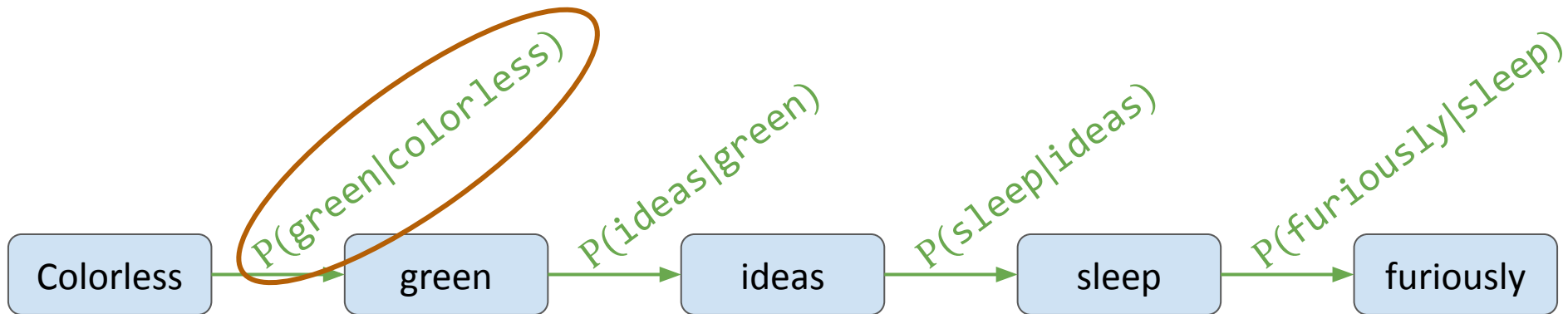
Visualizing a Markov language model

“Colorless green ideas sleep furiously.”



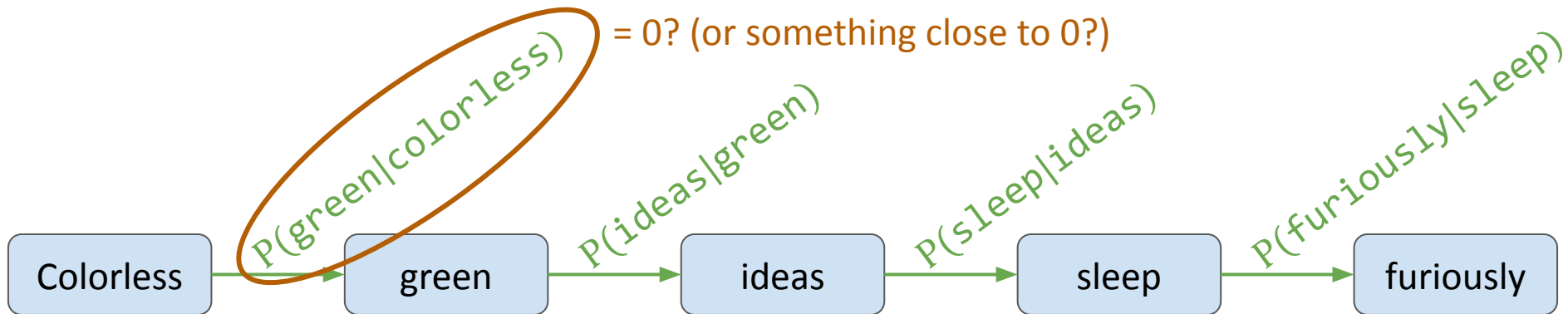
Visualizing a Markov language model

“Colorless green ideas sleep furiously.”



Visualizing a Markov language model

“Colorless green ideas sleep furiously.”



$P(\text{“Colorless green ideas sleep furiously”})$

$P(\text{“Berry berry berry berry berry”})$

$P(\text{“Colorless green ideas sleep furiously”})$

$> (?)$

$P(\text{“Berry berry berry berry berry”})$

adjective, adjective

$P(\text{"Colorless green ideas sleep furiously"})$

$> (?)$

$P(\text{"Berry berry berry berry berry"})$

noun, noun

adjective, adjective

$P(\text{“}\underline{\text{Colorless green}} \text{ ideas sleep furiously”})$

$> (?)$

$P(\text{“}\underline{\text{Berry berry}} \text{ berry berry berry”})$

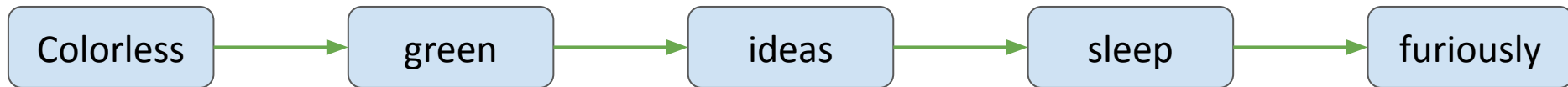
noun, noun

Maybe our human intuition is actually saying $P(JJ, JJ) > P(NN, NN)$?

Penn Treebank notation (see textbook): JJ = adjective, NN = singular noun

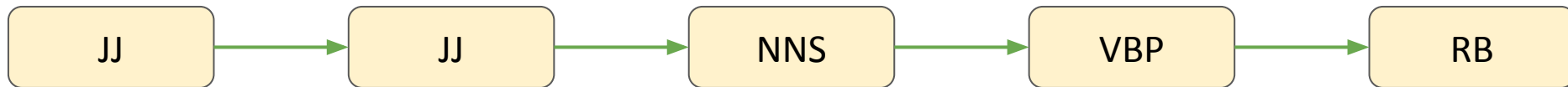
Refining our Markov model

“Colorless green ideas sleep furiously.”



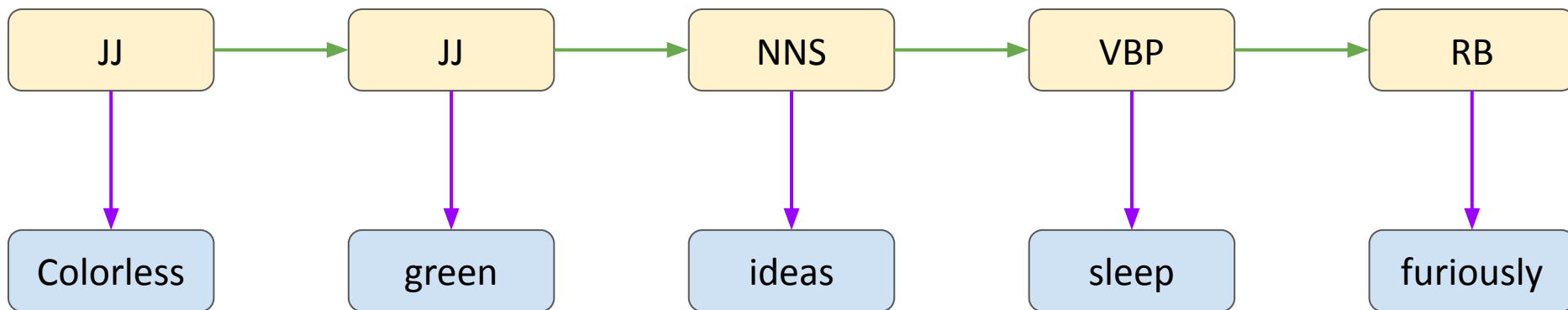
Refining our Markov model

“Colorless green ideas sleep furiously.”



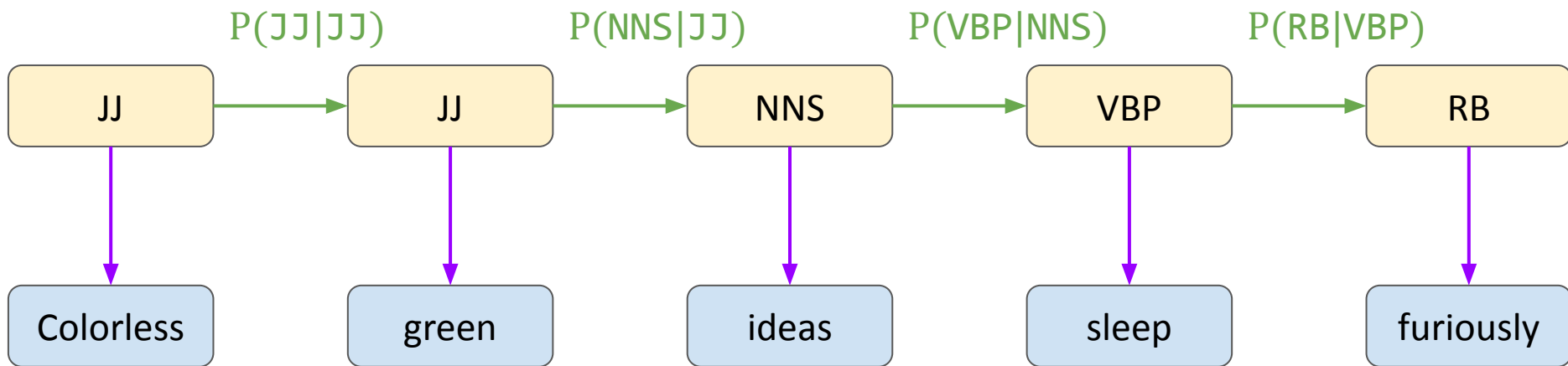
Refining our Markov model

“Colorless green ideas sleep furiously.”



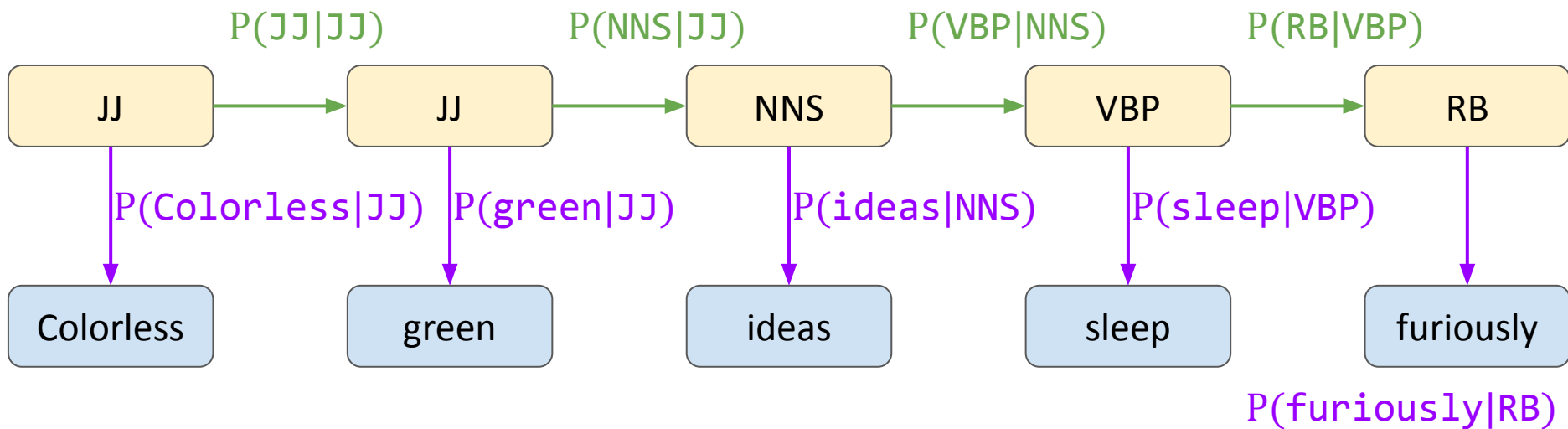
Refining our Markov model

“Colorless green ideas sleep furiously.”



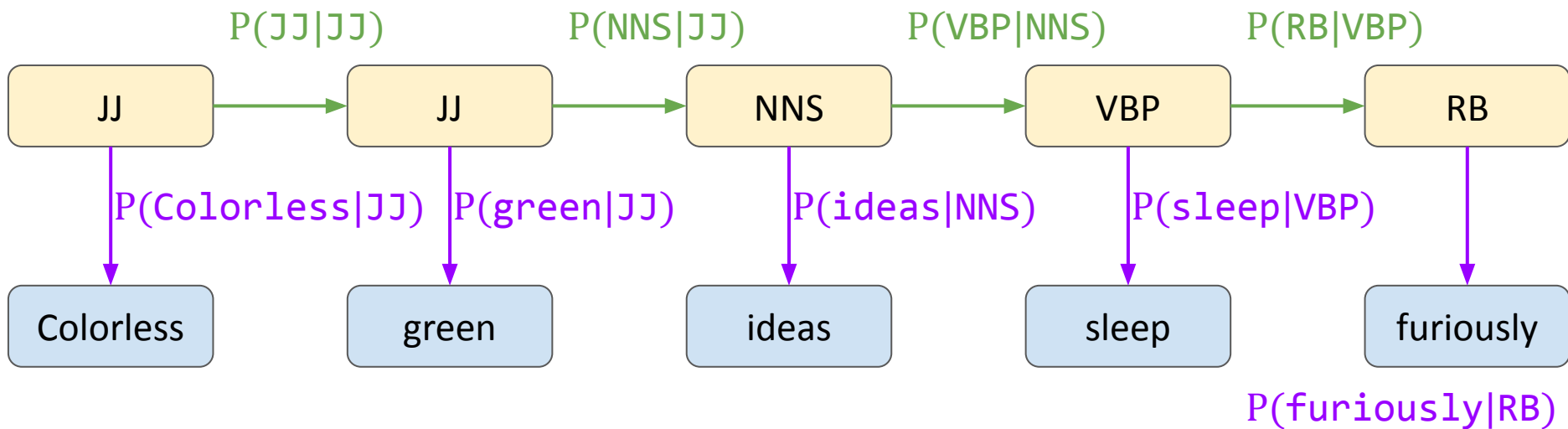
Refining our Markov model

“Colorless green ideas sleep furiously.”



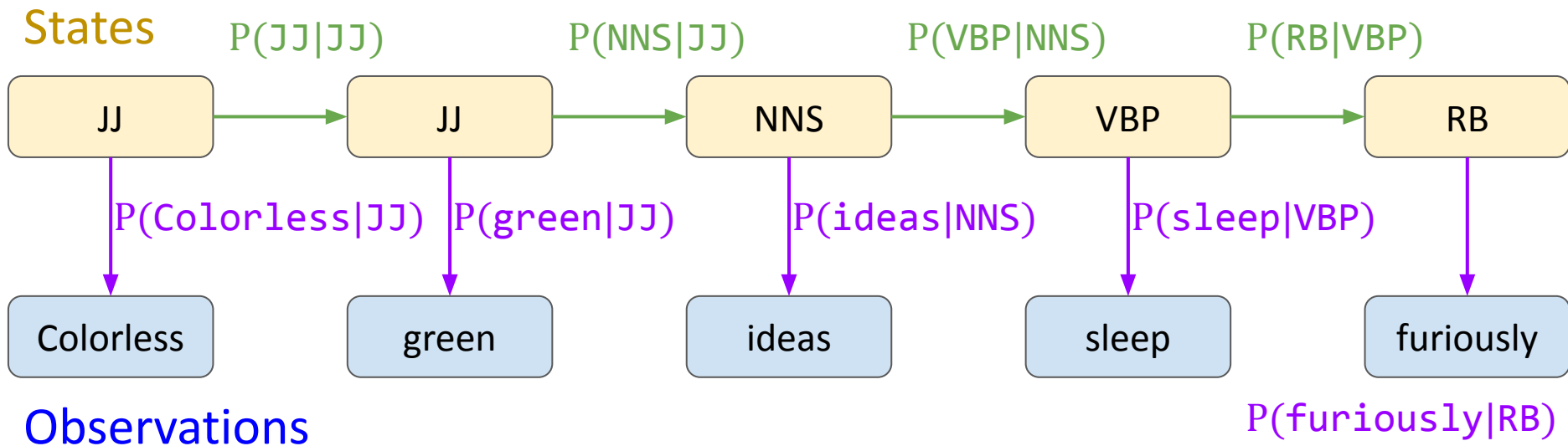
Hidden Markov Model

“Colorless green ideas sleep furiously.”



Hidden Markov Model

“Colorless green ideas sleep furiously.”



Hidden Markov Model

A Hidden Markov Model (HMM) is built upon *two* simplifying assumptions:

1. The Markov assumption (like before, but with part-of-speech tags instead of words):

$$P(t_1, t_2, t_3, \dots, t_k) \approx P(t_1)P(t_2|t_1)P(t_3|t_2)\dots P(t_k|t_{k-1})$$

2. The output independence assumption: generated words depend *only* on their part-of-speech tag

$$P(w_i|t_1, \dots, t_i, \dots, t_k, w_1, \dots, w_i, \dots, w_k) \approx P(w_i|t_i)$$

Applications of Hidden Markov Models

HMMs end up not being particularly great at generating text...

...but they *are* pretty good for a different task: **part-of-speech tagging** (aka POS tagging)

POS Tagging Formalized

Given a document:

$$d = w_1, w_2, w_3, \dots, w_k$$

POS Tagging Formalized

Given a document:

$$d = w_1, w_2, w_3, \dots, w_k$$

Want to find the most likely sequence of corresponding POS tags:

$$t_1, t_2, t_3, \dots, t_k$$

POS Tagging Formalized

Given a document:

$$d = w_1, w_2, w_3, \dots, w_k$$

Want to find the most likely sequence of corresponding POS tags:

$$t_1, t_2, t_3, \dots, t_k$$

Need to model:

$$P(t_1, t_2, t_3, \dots, t_k \mid w_1, w_2, w_3, \dots, w_k)$$

